3. Outer Measures
REAL VARIABLES I
Fall 2026

Our goal is to assign a meaningful notion of ”size” to subsets of a space X. The
An additive length cannot be defined on every subset of $\mathbb{R}$; a counterexample due to Vitali, later in these notes will make the obstruction precise. An *outer measure* is the useful first substitute. It is defined on all subsets, is built from countable coverings, and is countably subadditive: the size of a union is no greater than the sum of the sizes used to cover it.
After studying these notes, you will be able to do the following.

- Verify the defining properties of an outer measure.
- Generate an outer measure from a covering cost and determine when it extends that cost.
- Prove that Lebesgue outer measure agrees with length and area on rectangles.
- Explain why no translation-invariant length can be additive on every subset of $\mathbb{R}$.

3.1 Outer measures

Denoting by 2^X the collection of subsets of a set X, we would like to define a functional $\mu : 2^X \to \bar{\mathbb{R}}_+$ that has the following properties:

- **non-negative:** $\mu(E) \geq 0$ for all $E \subset X$;
- **zero on the empty set:** $\mu(\emptyset) = 0$;
- **monotone:** $\mu(A) \leq \mu(B)$ if $A \subset B$;
- **infinitely subadditive:** $\mu(\bigcup_{i\in\mathcal{J}} A_i) \leq \sum_{i\in\mathcal{J}} \mu(A_i)$;
- **infinitely additive on disjoint sets:** $\mu(\bigcup_{i\in\mathcal{J}} A_i) = \sum_{i\in\mathcal{J}} \mu(A_i)$ if $A_i \cap A_j = \emptyset$ whenever $i \neq j$.

We will see, however, that, generally, these conditions are too strong: in many settings there are no useful functionals that satisfy all the properties above. As a first restriction, we only consider countable subadditivity and additivity, also known as σ-(sub)additivity.
Even then, these conditions are generally too restrictive, which motivates us to require fewer properties, thereby leading to the notion of an *outer measure*.

Definition 3.1.1 (Outer measure). An *outer measure* on X is a function $\mu^* : 2^X \to [0, \infty]$ with the following properties:

1. $\mu^*(\emptyset) = 0$;
2. **monotonicity:** if $E \subseteq F \subseteq X$, then $\mu^*(E) \leq \mu^*(F)$;
3. **σ-subadditivity:** for every sequence $(E_n \in 2^X)_{n\in\mathbb{N}}$ it holds $\mu^*\left(\bigcup_{n\in\mathbb{N}} E_n\right) \leq \sum_{n\in\mathbb{N}} \mu^*(E_n)$;

The third property is called *countable subadditivity* or *σ-subadditivity*.

An outer measure is defined on all subsets of X, and the sets involved in the σ-subadditivity property need not be disjoint. For this reason we require only the inequality $\leq$ there. Monotonicity is an additional property; it does not follow from σ-subadditivity.

Example 3.1.2 (Counting outer measure). For $E \subseteq X$, set

$$\mu^*(E) \coloneqq \begin{cases} \# E & \text{if } E \text{ is finite,} \\ \infty & \text{otherwise.} \end{cases}$$

Let us show that μ^* is an outer measure. The definition gives $\mu^*(\emptyset) = 0$, and $E \subseteq F$ implies $\mu^*(E) \leq \mu^*(F)$. To prove countable subadditivity, let $(E_n)_{n\in\mathbb{N}}$ be a sequence of subsets of X. If $\sum_{n\in\mathbb{N}} \mu^*(E_n) < \infty$, then only finitely many of the sets E_n are nonempty, and each of them is finite. Hence

$$\mu^*\left(\bigcup_{n\in\mathbb{N}} E_n\right) = \#\left(\bigcup_{n\in\mathbb{N}} E_n\right) \leq \sum_{n\in\mathbb{N}} \# E_n = \sum_{n\in\mathbb{N}} \mu^*(E_n).$$

If the sum on the right is infinite, the desired inequality is automatic.
A slightly more involved example is a "weighted" version of the above construction.

Example 3.1.3 (Weighted counting outer measure). Let $w\colon X \to [0,\infty]$. For $E \subseteq X$, define

$$\mu_w^*(E) \coloneqq \sum_{x\in E} w(x) \coloneqq \sup\left\{\sum_{x\in F} w(x) \;:\; F \subseteq E, F \text{ finite}\right\}. \tag{1}$$

Then μ_w^* is an outer measure and is countably additive on disjoint sequences. Counting outer measure is the case $w(x) = 1$ for every $x \in X$.
Indeed, the definition immediately gives $\mu_w^*(\emptyset) = 0$ and monotonicity. Let $(E_n)_{n\in\mathbb{N}}$ be any sequence of subsets of X. If $F \subseteq \bigcup_{n\in\mathbb{N}} E_n$ is finite, assign to every $x \in F$ one index $n(x)$ for which $x \in E_{n(x)}$. Then

$$\sum_{x\in F} w(x) = \sum_{n\in\mathbb{N}} \sum_{\substack{x\in F \\ n(x)=n}} w(x) \leq \sum_{n\in\mathbb{N}} \mu_w^*(E_n).$$

Taking the supremum over all finite F proves countable subadditivity.
Now suppose that the sets E_n are pairwise disjoint. The inequality already proved gives

$$\mu_w^*\left(\bigcup_{n\in\mathbb{N}} E_n\right) \leq \sum_{n\in\mathbb{N}} \mu_w^*(E_n).$$

For the reverse inequality, finite additivity of the sums in (1) follows by partitioning each finite subset according to the disjoint sets containing its points. Hence, for every finite $J \subseteq \mathbb{N}$,

$$\sum_{n\in J} \mu_w^*(E_n) = \mu_w^*\left(\bigcup_{n\in J} E_n\right) \leq \mu_w^*\left(\bigcup_{n\in\mathbb{N}} E_n\right).$$

Taking the supremum over finite J proves countable additivity.

Example 3.1.4 (Lebesgue outer measure). For any $E \subset \mathbb{R}$ define

$$\mathcal{L}_1^*(E) \coloneqq \inf\Big\{\sum_{j=0}^{\infty}(b_j - a_j) \;:\; \bigcup_{j=0}^{\infty}(a_j, b_j) \supseteq E\Big\}.$$

Next, we will see that the formula used to define $\mathcal{L}_1^*$ is a common way to define outer measures. The fact that $\mathcal{L}_1^*$ is indeed an outer measure will follow from general properties of this construction.
The first two examples are actually measures: they satisfy countable additivity on disjoint families. In contrast, we will conclude these notes by showing that Lebesgue outer measure is not even finitely additive on all subsets.

3.1.1 Outer measures generated by costs

Next, we illustrate a very common way to explicitly define outer measures.
Let $\mathcal{A} \subseteq 2^X$ contain $\emptyset$, and let $\ell\colon \mathcal{A} \to [0,\infty]$ satisfy $\ell(\emptyset) = 0$. We regard $\ell(A)$ as the "cost" of using $A \in \mathcal{A}$ in a cover of a set E.

Definition 3.1.5 (Outer measure generated by a cost)**.** For $E \subseteq X$, define the *outer measure generated by* ℓ as

$$\ell^*(E) \coloneqq \inf\left\{\sum_{n\in\mathbb{N}} \ell(A_n) \,:\, (A_n \in \mathcal{A})_{n\in\mathbb{N}}, \quad E \subseteq \bigcup_{n\in\mathbb{N}} A_n\right\}. \tag{2}$$

If no countable family in $\mathcal{A}$ covers E, the infimum is ∞.

Let us show that the functional ℓ^* in (2) is an outer measure on X.

Proof. The empty-set and monotonicity axioms follow directly from the covering definition. For subadditivity, we combine a nearly optimal cover of each set into one countable cover of their union.

Step 1: the empty set and monotonicity.

The constant sequence $A_n = \emptyset$ covers $\emptyset$ and has cost zero. Hence $\ell^*(\emptyset) = 0$.
If $E \subseteq F$, then every countable cover of F is a countable cover of E. Taking infima gives $\ell^*(E) \leq \ell^*(F)$.

Step 2: countable subadditivity.

It remains to prove countable subadditivity. Let $(E_n \in 2^X)_{n\in\mathbb{N}}$. If $\ell^*(E_n) = \infty$ for some n, the required inequality is immediate. Assume that $\ell^*(E_n) < \infty$ for every n, and fix $\varepsilon > 0$. For each $n \in \mathbb{N}$, choose a sequence $(A_{n,k} \in \mathcal{A})_{k\in\mathbb{N}}$ such that

$$E_n \subseteq \bigcup_{k\in\mathbb{N}} A_{n,k} \quad \text{and} \quad \sum_{k\in\mathbb{N}} \ell(A_{n,k}) \leq \ell^*(E_n) + \varepsilon 2^{-n-1}.$$

The family $(A_{n,k})_{(n,k)\in\mathbb{N}^2}$ is countable and covers $\bigcup_{n\in\mathbb{N}} E_n$. Therefore

$$\ell^*\left(\bigcup_{n\in\mathbb{N}} E_n\right) \leq \sum_{n\in\mathbb{N}}\sum_{k\in\mathbb{N}} \ell(A_{n,k}) \leq \sum_{n\in\mathbb{N}} \ell^*(E_n) + \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, this proves countable subadditivity. □

The construction has the following extremal property: ℓ^* is the *largest* outer measure that is smaller than ℓ on $\mathcal{A}$.

Proposition 3.1.6 (Largest outer measure below a cost)**.** For every $A \in \mathcal{A}$, $\ell^*(A) \leq \ell(A)$.
Moreover, if ν^* is an outer measure on X such that $\nu^*(A) \leq \ell(A)$ for every $A \in \mathcal{A}$, then $\nu^*(E) \leq \ell^*(E)$ for every $E \subseteq X$.

Proof. (proves 3.1.6)
The proof compares every eligible cover with the two outer measures. To show that $\ell^*(A) \leq \ell(A)$, note that the set A together with countably many copies of $\emptyset$ is a countable cover of A.
Next, suppose $\nu^*(A) \leq \ell(A)$ for every $A \in \mathcal{A}$. Fix $E \subseteq X$ and let us show that $\nu^*(E) \leq \ell^*(E)$. For any countable cover $(A_n \in \mathcal{A})_{n\in\mathbb{N}}$ of E, monotonicity and countable subadditivity give

$$\nu^*(E) \leq \nu^*\left(\bigcup_{n\in\mathbb{N}} A_n\right) \leq \sum_{n\in\mathbb{N}} \nu^*(A_n) \leq \sum_{n\in\mathbb{N}} \ell(A_n).$$

Taking the infimum over all such covers proves Proposition 3.1.6. □

Example 3.1.7 (A generated outer measure larger than weighted counting)**.** Enumerate $\mathbb{Q}$ as $(q_n)_{n\geq 1}$, and define $\ell\colon 2^{\mathbb{R}} \to [0,\infty]$ by

$$\ell(E) = \begin{cases} \sum_{q_n \in E} 2^{-n}, & E \text{ is countable,} \\ \infty, & E \text{ is uncountable.} \end{cases}$$

Also define the weighted counting outer measure

$$\nu^*(E) = \sum_{q_n \in E} 2^{-n}.$$

Then $\nu^*(E) \leq \ell(E)$ for every $E \subseteq \mathbb{R}$, so Proposition 3.1.6 gives $\nu^* \leq \ell^*$.
If E is countable, the one-set cover of E gives $\ell^*(E) \leq \ell(E) = \nu^*(E)$. Hence $\ell^*(E) = \nu^*(E)$. If E is uncountable, every countable cover of E contains an uncountable set. Every such cover has infinite cost, and therefore $\ell^*(E) = \infty$. Consequently,

$$\ell^*(E) = \begin{cases} \sum_{q_n \in E} 2^{-n}, & E \text{ is countable,} \\ \infty, & E \text{ is uncountable.} \end{cases}$$

In particular, ℓ^* is strictly larger than ν^* on every uncountable subset of $\mathbb{R}$.

In general, according to (2), it can happen that $\ell^*(A) < \ell(A)$: there is a way to cover a set $A \in \mathcal{A}$ more efficiently than the cost functional ℓ describes. Mostly, however, we are interested in using ℓ^* to *extend* ℓ. The following condition precisely characterizes when ℓ^* coincides with ℓ on the original collection $\mathcal{A}$.

Proposition 3.1.8 (Extension of a cost)**.** The identity

$$\ell^*(A) = \ell(A), \qquad A \in \mathcal{A},$$

holds if and only if the following covering inequality holds: for every $A \in \mathcal{A}$ and every sequence $(A_n \in \mathcal{A})_{n\in\mathbb{N}}$,

$$A \subseteq \bigcup_{n\in\mathbb{N}} A_n \quad \Longrightarrow \quad \ell(A) \leq \sum_{n\in\mathbb{N}} \ell(A_n). \tag{3}$$

When $\mathcal{A} = 2^X$, condition (3) is equivalent to monotonicity and countable subadditivity of ℓ.

Proof. *(proves 3.1.8)*
Left to the reader □

3.1.2 Lebesgue outer measure in dimensions one and two

In this section we focus on the Lebesgue outer measure on $\mathbb{R}^d$. For simplicity, we will focus on the case $d = 1$ and $d = 2$. Higher dimensional cases are notationally more complex but present no additional difficulties. An open rectangle in $\mathbb{R}^d$ is a set of the form

$$\prod_{j=1}^{d} (a_j, b_j), \qquad -\infty < a_j < b_j < \infty,$$

and its volume is $\mathrm{Vol}_d\left(\prod_{j=1}^{d}(a_j, b_j)\right) = \prod_{j=1}^{d}(b_j - a_j)$.

Definition 3.1.9 (Lebesgue outer measure). For $E \subseteq \mathbb{R}^d$, define

$$\mathcal{L}_d^*(E) \coloneqq \inf\left\{\sum_{n\in\mathbb{N}} \mathrm{Vol}_d(R_n) : (R_n)_{n\in\mathbb{N}} \text{ are open rectangles and } E \subseteq \bigcup_{n\in\mathbb{N}} R_n\right\}. \tag{4}$$

The definition uses the construction in Definition 3.1.5. It is not yet automatic that a rectangle has outer measure equal to its usual length or area.

Proposition 3.1.10 (Lebesgue outer measure). For $d \in \{1, 2\}$, the functional $\mathcal{L}_d^*$ is an outer measure on $\mathbb{R}^d$, and $\mathcal{L}_d^*(R) = \mathrm{Vol}_d(R)$ for all open rectangles.

Proof. (proves 3.1.10)
The proof first reduces an arbitrary countable cover to a finite cover of a compact inner rectangle. A finite grid then supplies the area comparison.

Step 1: reduce to a finite cover of a compact rectangle.

The fact that $\mathcal{L}_d^*$ is an outer measure follows from the generated-cost construction above, applied to the family consisting of the open rectangles and $\emptyset$, with cost Vol_d.
We verify that $\mathcal{L}_d^*(R) = \mathrm{Vol}_d(R)$ by checking, according to Proposition 3.1.8, that for any open rectangle R it holds that

$$\mathrm{Vol}_d(R) \le \sum_{n=1}^{\infty} \mathrm{Vol}_d(R_n)$$

whenever $R \subseteq \bigcup_{n=1}^{\infty} R_n$. For simplicity, we present the proof for $d = 2$.
Let $R = (A_1, B_1) \times (A_2, B_2)$. It is sufficient to prove that

$$\sum_{n=1}^{\infty} \mathrm{Vol}_2(R_n) \ge (B_1' - A_1')(B_2' - A_2')$$

whenever

$$K = [A_1', B_1'] \times [A_2', B_2'] \subset R,$$

because the areas of such inner rectangles increase to $\mathrm{Vol}_2(R)$. We make this reduction in order to use compactness.
Since $(R_n)_{n\in\mathbb{N}}$ is an open cover of the compact set K, it has a finite subcover. After relabeling this subcover, we may write

$$K \subseteq \bigcup_{n=1}^{N} R_n.$$

Thus it remains to prove the covering inequality for a finite family of rectangles.

Step 2: compare areas on a finite grid.

Draw all vertical and horizontal lines containing sides of K or of one of $R_1, \dots, R_N$. These lines refine K into finitely many grid rectangles. The interior of each grid rectangle is contained in at least one R_n. Indeed, choose a point of that interior and an R_n containing it; since every side of R_n lies on one of the grid lines, no side of R_n crosses the interior of the grid rectangle. Assign the grid rectangle to one such R_n.
We now use induction on the number of grid lines. Before any line is inserted, the total area is the area of the original rectangle. Inserting one new line replaces every rectangle it crosses by two rectangles, and their areas add to the area of the original one. Consequently, any such rectangular refinement preserves total area.

It follows that the grid rectangles have total area $\mathrm{Vol}_2(K)$. For each n, refine R_n by the same grid lines. The preceding induction shows that the total area of all pieces of R_n is $\mathrm{Vol}_2(R_n)$; hence the grid rectangles assigned to R_n have total area at most $\mathrm{Vol}_2(R_n)$. Summing over n gives

$$\mathrm{Vol}_2(K) \le \sum_{n=1}^{N} \mathrm{Vol}_2(R_n) \le \sum_{n=1}^{\infty} \mathrm{Vol}_2(R_n).$$

Letting $A_i' \downarrow A_i$ and $B_i' \uparrow B_i$ proves the required covering inequality for R. The proof for $d = 1$ is identical, with division points in place of vertical and horizontal lines. Therefore Proposition 3.1.8 gives

$$\mathcal{L}_d^*(R) = \mathrm{Vol}_d(R), \qquad d \in \{1, 2\}.$$

□

3.1.3 The obstruction to additivity on all subsets

The construction of the Lebesgue outer measure $\mathcal{L}_1^*$ on $\mathbb{R}$ described above provides a translation-invariant functional on $2^{\mathbb{R}}$ that coincides with length on open intervals. We would also like it to satisfy $\mathcal{L}_1^*(E \cup F) = \mathcal{L}_1^*(E) + \mathcal{L}_1^*(F)$ for any two disjoint sets $E, F \subset \mathbb{R}$. The Vitali counterexample below shows that this is not the case and, more generally, that there is no countably additive translation-invariant positive functional on $2^{\mathbb{R}}$ that assigns every bounded interval its length.

Lemma 3.1.11 (Translation invariance)**.** For every $E \subseteq \mathbb{R}$ and $t \in \mathbb{R}$ it holds that $\mathcal{L}_1^*(E + t) = \mathcal{L}_1^*(E)$.

Proof. (proves 3.1.11)
Translating an open interval cover of E by t gives an open interval cover of $E + t$ with the same total cost. Hence $\mathcal{L}_1^*(E + t) \le \mathcal{L}_1^*(E)$. Applying this inequality with $E + t$ and $-t$ gives the reverse inequality. □

Proposition 3.1.12 (Vitali obstruction)**.** Lebesgue outer measure is not countably additive on $2^{\mathbb{R}}$. More generally, there is no countably additive function $\mu \colon 2^{\mathbb{R}} \to [0, \infty]$ that is translation invariant and assigns to $[0, 1]$ a finite non-zero quantity.

First, we note that a σ-additive μ is automatically monotone: If $A \subset B$ then $\mu(B) = \mu(A \cup (B \setminus A)) = \mu(A) + \mu(B \setminus A) \ge \mu(A)$.

Proof. (proves 3.1.12)
The proof constructs countably many disjoint translates of one representative set and compares their measure with the lengths of an interval

Step 1: construct disjoint rational translates.

On $[0, 1]$, define $x \sim y \iff x - y \in \mathbb{Q}$. This is an equivalence relation.
Next we want to choose a set $V \subseteq (0, 1/2)$ containing exactly one representative of each equivalence class. The axiom of choice allows us to choose $V_0 \subseteq \mathbb{R}$ containing exactly one representative of each equivalence class. To construct V, use the axiom of choice againt to find for each $z \in V_0$ a rational $q_z \in (z - 1/2, z)$. Next define

$$V \coloneqq \{z - q_z \,:\, z \in V_0\}.$$

Clearly, $V \subset (0, 1/2)$ by construction. Also, V contains a representative of each equivalence class once because $z - q_z \sim z$ for each $z \in V_0$.
Next, let us discuss two cases $\mu(V) = 0$ or $\mu(V) > 0$.

Step 2: Case $\mu(V) = 0$

Note that

$$[0,1] \subset \mathbb{R} = \bigcup_{q\in\mathbb{Q}} V + q.$$

Note that for $q \neq q'$ it holds that $(V+q) \cap (V+q') = \emptyset$, therefore the union above is of pairwise disjoint sets. The union is also indexed by a countable index set. It follows that

$$\mu([0,1]) \leq \mu(\bigcup_{q\in\mathbb{Q}} V + q) = \sum_{q\in\mathbb{Q}} \mu(V+q) = \sum_{q\in\mathbb{Q}} \mu(V) = \sum_{q\in\mathbb{Q}} 0 = 0.$$

Step 2: Case $\mu(V) > 00$

Note that for any $q \in (0,1/2)$we have that $V + q \subset (0,1)$ so

$$[0,1] \supset \bigcup_{q\in(0,1/2)\cap\mathbb{Q}} V + q.$$

Note that for $q \neq q'$ it holds that $(V+q) \cap (V+q') = \emptyset$, therefore the union above is of pairwise disjoint sets. The union is also indexed by a countable index set. It follows that

$$\mu([0,1]) \geq \mu(\bigcup_{q\in(0,1/2)\cap\mathbb{Q}} V + q) = \sum_{q\in(0,1/2)\cap\mathbb{Q}} \mu(V+q) = \sum_{q\in(0,1/2)\cap\mathbb{Q}} \mu(V) = +\infty \cdot \mu(V) = +\infty.$$

□

Corollary 3.1.13 (Failure of finite additivity). There are disjoint sets $E, F \subseteq \mathbb{R}$ such that

$$\mathcal{L}_1^*(E \cup F) \neq \mathcal{L}_1^*(E) + \mathcal{L}_1^*(F).$$

Thus, although Lebesgue outer measure is countably subadditive, it is not additive even on pairs of disjoint sets.

Proof. (proves 3.1.13)

Pairwise additivity would imply finite additivity. Together with the existing countable subadditivity, it would then imply countable additivity.

Step 1: obtain finite additivity.

Suppose instead that $\mathcal{L}_1^*$ were additive on every pair of disjoint sets. Then it would be additive on every finite disjoint family: if $E_1, \dots, E_N$ are pairwise disjoint, repeated use of the assumed pairwise additivity gives

$$\mathcal{L}_1^*\left(\bigcup_{n=1}^{N} E_n\right) = \sum_{n=1}^{N} \mathcal{L}_1^*(E_n).$$

Step 2: pass from finite to countable additivity.

Now let $(E_n)_{n\in\mathbb{N}}$ be any pairwise disjoint sequence and put $E = \bigcup_{n\in\mathbb{N}} E_n$. Monotonicity gives

$$\sum_{n=1}^{N} \mathcal{L}_1^*(E_n) = \mathcal{L}_1^*\left(\bigcup_{n=1}^{N} E_n\right) \leq \mathcal{L}_1^*(E) \qquad \text{for every } N.$$

Taking the supremum over N, and using countable subadditivity for the reverse inequality, yields

$$\mathcal{L}_1^*(E) = \sum_{n\in\mathbb{N}} \mathcal{L}_1^*(E_n).$$

This would make $\mathcal{L}_1^*$ countably additive on $2^{\mathbb{R}}$, contradicting Proposition 3.1.12.

$\square$